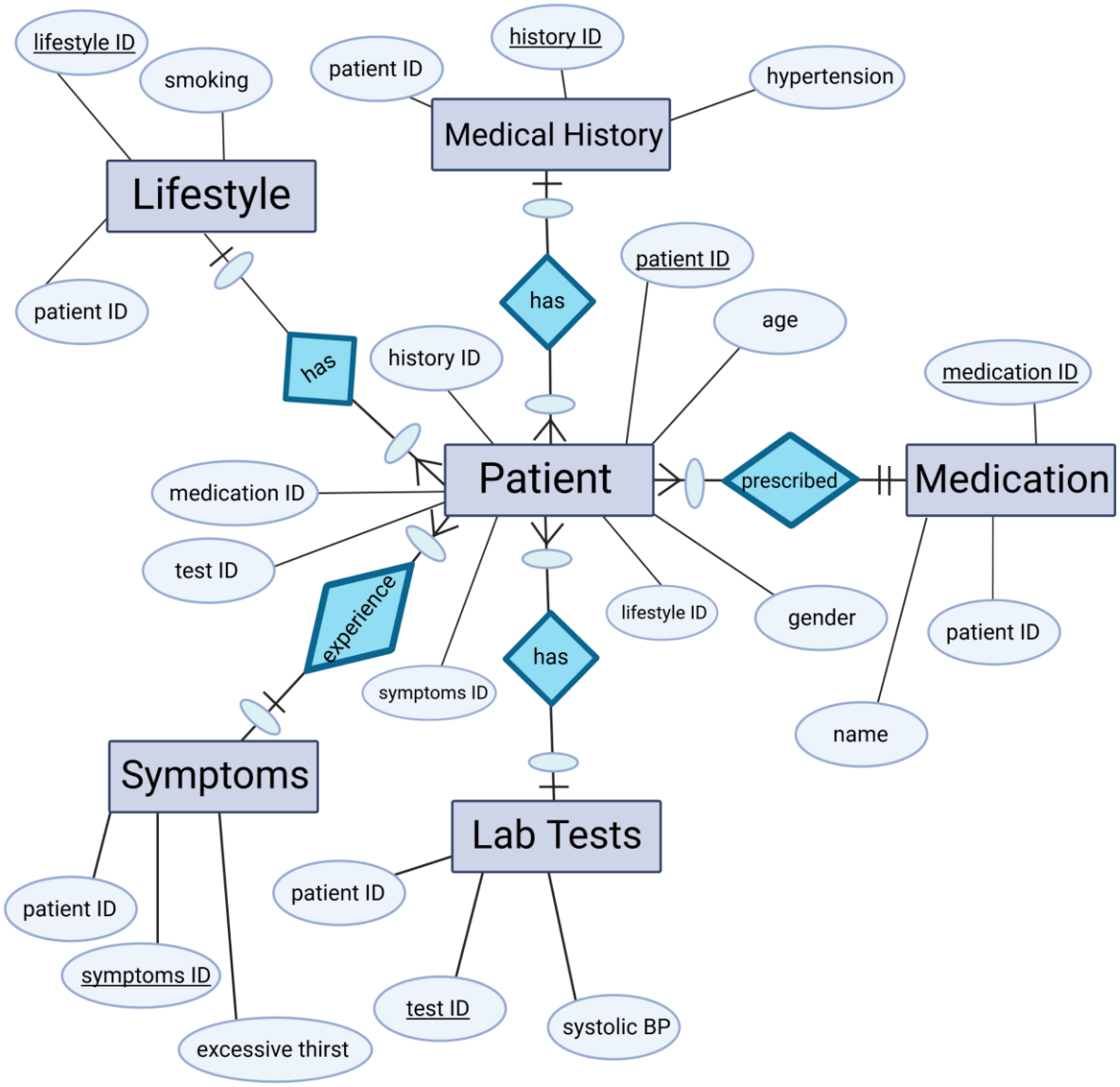


Mechanisms of Diabetes

Objective of creating this ER diagram and the application of this diagram

The ER diagram is used as a blueprint for creating the database structure. It guides the creation of tables, defining primary keys, foreign keys, and relationships between entities. Each of these entities becomes a table in the database, with attributes represented as columns. The diagram makes it easier to understand the relationships between different pieces of data. For example, if you need to pull up all lab test results for a specific patient, the diagram shows you how to link the Patient table to the LabTests table using the PatientID, making the process smooth and efficient.

The primary goal of this ER diagram is to organize and structure patient healthcare data efficiently. It allows the organization of various aspects of patient care, such as lifestyle

habits (e.g., smoking), symptoms, medical history (e.g., hypertension), lab test results (e.g., systolic BP), and prescribed medications. This ER diagram is used to guide the creation of a relational database for storing patient-related information in a well-organized manner. The entities in the diagram—such as Patient, Lifestyle, Symptoms, MedicalHistory, LabTests, and Medication—become the core tables in the database, with their relationships determining how the data will be linked and accessed.

Proved of 3NF

To ensure the database design is in Third Normal Form (3NF), the database must meet 2 requirement which must be already in the Second Normal Form(2NF) and without the transitive dependencies between non-key attributes and the primary key.

In this database, each table focus on the patient information such as lifestyle, symptoms, medical history, lab tests and medications. These tables are liked via the foreign keys to make sure data remains consistent and related. For instance, the Patient table references other tables like Lifestyle, Symptoms, LabTests, MedicalHistory, and Medication via foreign keys (LifestyleID, SymptomsID, LabTestsID, HistoryID, MedicationID). This ensuring the data store only once in their appropriate tables and the Patient table holds only the references (IDs) to avoid data redundancy.

Moreover, this database also satisfied the requirements which must not contain any transitive dependencies. Non-key attributes such as Age, Gender, or SystolicBP depend solely on the primary key (e.g., PatientID or LabTestsID) rather than on other non-key attributes. For example, in the LabTests table, SystolicBP is tied directly to the LabTestsID, and in the MedicalHistory table, Hypertension is linked to the HistoryID. This approach helps avoid repeating the same information in multiple places, keeping the data neat and easy to understand. By sticking to the principles of 3NF, the structure not only makes the database more organized but also simplifies the process of managing, accessing, and updating the data. It ensures everything stays consistent and reduces the risk of mistakes or mismatches in the data.

Use cases

1.

```sql
--- Use Case 1: Show the details of all patients with their lifestyle, symptoms, medical history, lab tests, and medication
SELECT p.PatientID, p.Age, p.Gender,
       l.Smoking AS SmokingStatus,
       s.ExcessiveThirst AS Symptoms,
       mh.Hypertension AS MedicalHistory,
       lt.SystolicBP AS LabTestSystolicBP,
       m.MedicationName AS Medication
FROM Patient p
JOIN Lifestyle l ON p.LifestyleID = l.LifestyleID
JOIN Symptoms s ON p.SymptomsID = s.SymptomsID
JOIN MedicalHistory mh ON p.HistoryID = mh.HistoryID
JOIN LabTests lt ON p.LabTestsID = lt.LabTestsID
JOIN Medication m ON p.MedicationID = m.MedicationID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PatientID	Age	Gender	SmokingStatus	Symptoms	MedicalHistory	LabTestSystolicBP	Medication
6205	56	Male	NO	YES	NO	136.00	Statins
6208	48	Female	YES	NO	NO	123.00	Statins
6209	90	Male	NO	YES	NO	169.00	Statins
6200	31	Male	YES	NO	YES	92.00	Statins
6201	30	Female	NO	NO	NO	90.00	AntihypertensiveMedications
6202	45	Male	NO	NO	NO	164.00	AntihypertensiveMedications
6210	66	Male	NO	YES	NO	143.00	AntihypertensiveMedications
6203	86	Female	NO	NO	YES	144.00	AntihypertensiveMedications
6206	75	Female	YES	YES	YES	91.00	AntihypertensiveMedications
6207	46	Male	NO	YES	YES	138.00	AntihypertensiveMedications

2.

```sql
--- Use Case 2: Show patients with a specific medical history (e.g., hypertension)
SELECT p.PatientID, p.Age, p.Gender,
       l.Smoking AS SmokingStatus,
       s.ExcessiveThirst AS Symptoms,
       mh.Hypertension AS MedicalHistory,
       lt.SystolicBP AS LabTestSystolicBP,
       m.MedicationName AS Medication
FROM Patient p
JOIN Lifestyle l ON p.LifestyleID = l.LifestyleID
JOIN Symptoms s ON p.SymptomsID = s.SymptomsID
JOIN MedicalHistory mh ON p.HistoryID = mh.HistoryID
JOIN LabTests lt ON p.LabTestsID = lt.LabTestsID
JOIN Medication m ON p.MedicationID = m.MedicationID
WHERE mh.Hypertension = 'YES';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PatientID	Age	Gender	SmokingStatus	Symptoms	MedicalHistory	LabTestSystolicBP	Medication
6200	31	Male	YES	NO	YES	92.00	Statins
6203	86	Female	NO	NO	YES	144.00	AntihypertensiveMedications
6206	75	Female	YES	YES	YES	91.00	AntihypertensiveMedications
6207	46	Male	NO	YES	YES	138.00	AntihypertensiveMedications

3.

```
30  --- Use Case 3: Show patients who are prescribed a specific medication (e.g., Statins)
31  SELECT p.PatientID, p.Age, p.Gender,
32         l.Smoking AS SmokingStatus,
33         s.ExcessiveThirst AS Symptoms,
34         mh.Hypertension AS MedicalHistory,
35         lt.SystolicBP AS LabTestSystolicBP,
36         m.MedicationName AS Medication
37  FROM Patient p
38  JOIN Lifestyle l ON p.LifestyleID = l.LifestyleID
39  JOIN Symptoms s ON p.SymptomsID = s.SymptomsID
40  JOIN MedicalHistory mh ON p.HistoryID = mh.HistoryID
41  JOIN LabTests lt ON p.LabTestsID = lt.LabTestsID
42  JOIN Medication m ON p.MedicationID = m.MedicationID
43  WHERE m.MedicationName = 'Statins';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PatientID	Age	Gender	SmokingStatus	Symptoms	MedicalHistory	LabTestSystolicBP	Medication
6200	31	Male	YES	NO	YES	92.00	Statins
6205	56	Male	NO	YES	NO	136.00	Statins
6208	48	Female	YES	NO	NO	123.00	Statins
6209	90	Male	NO	YES	NO	169.00	Statins

4.

```
45  --- Use Case 4: Show patients based on a certain age range (e.g., patients aged 40-60)
46  SELECT p.PatientID, p.Age, p.Gender,
47         l.Smoking AS SmokingStatus,
48         s.ExcessiveThirst AS Symptoms,
49         mh.Hypertension AS MedicalHistory,
50         lt.SystolicBP AS LabTestSystolicBP,
51         m.MedicationName AS Medication
52  FROM Patient p
53  JOIN Lifestyle l ON p.LifestyleID = l.LifestyleID
54  JOIN Symptoms s ON p.SymptomsID = s.SymptomsID
55  JOIN MedicalHistory mh ON p.HistoryID = mh.HistoryID
56  JOIN LabTests lt ON p.LabTestsID = lt.LabTestsID
57  JOIN Medication m ON p.MedicationID = m.MedicationID
58  WHERE p.Age BETWEEN 40 AND 60;
59
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PatientID	Age	Gender	SmokingStatus	Symptoms	MedicalHistory	LabTestSystolicBP	Medication
6200	31	Male	YES	NO	YES	92.00	Statins
6205	56	Male	NO	YES	NO	136.00	Statins
6208	48	Female	YES	NO	NO	123.00	Statins
6209	90	Male	NO	YES	NO	169.00	Statins

Reference:

综合糖尿病健康数据集 - Heywhale.com (no date). Available at: https://www.heywhale.com/mw/dataset/666a5579c32ba1f3df7b7e02/file (Accessed: 5 March 2025).